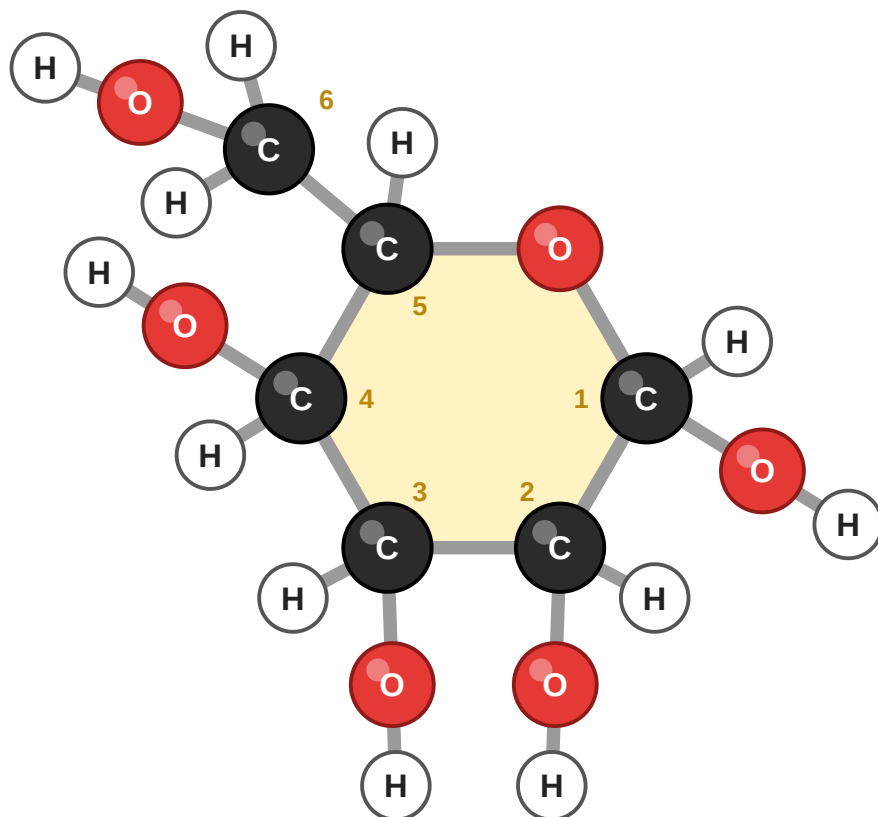


Molecule Builder

Bonus: Sugar (glucose)

The sugar that powers your body. It is a ring of 6 with lots of little O-H "handles."



6 carbon · 12 hydrogen · 6 oxygen
24 atoms in one molecule

Parts you need

- **6** Carbon (black)
- **6** Oxygen (red)
- **12** Hydrogen (white)
- **24** straight sticks

No bendy sticks: every bond is single.

How heavy?

$$6 \text{ C} \times 12 = 72$$

$$12 \text{ H} \times 1 = 12$$

$$6 \text{ O} \times 16 = 96$$

$$\text{Total} = 180$$

As heavy as 180 hydrogen atoms,
and 10× heavier than water (18).

1 Make the ring

Connect **5 carbons + 1 oxygen** in a loop of 6 (6 sticks). The oxygen sits in the ring like one of the corners. This is the shaded hexagon.



2 Number the carbons

Start at the carbon right next to the ring oxygen and count around: 1, 2, 3, 4, 5. The small gold numbers in the picture match.



3 Add the arm

On carbon 5, add one more carbon sticking out. That is **carbon 6**.



4 Add the OH groups

Build 5 little "O-H" pieces (1 red + 1 white each). Plug one onto carbons 1, 2, 3, 4 and 6.



5 Cap it with hydrogens

Every carbon needs 4 sticks. Fill each empty hole with a white hydrogen: carbons 1 to 5 get 1 each, carbon 6 gets 2. That uses 7.



6 Check your count

You should have used **6 black, 6 red, 12 white** and **24 sticks**. Every carbon has 4 sticks, every oxygen 2, every hydrogen 1.



Plants make it

Leaves take **6 CO₂ + 6 H₂O** plus sunlight and build 1 glucose, letting out 6 O₂. Those are molecules from your first page!

You burn it

Your body runs that backward: glucose + oxygen → CO₂ + water + **energy**. That is why you breathe out CO₂.

Why it dissolves

All those O-H handles make sugar sticky to water, just like water sticks to itself. So water pulls sugar apart and it disappears in your tea.